

M.S. PROJECT PROPOSAL

“Determinants of Financial Worries in SAARC Countries: An Expanded Multinomial and Machine-Learning Approach with Country and Life-Cycle Effects”

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University of Chittagong

Submitted To: Professor Mr. Md. Monirul Islam

Department of Statistics, University of Chittagong

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Dear Sir,

With due respect, I would like to submit the following proposal, titled above, for your kind review and approval, in partial fulfillment of the requirements for the degree of Master of Science (M.S.) in Statistics. I would be grateful for your guidance to proceed with this research under your supervision.

Yours sincerely

Md. Mahbub Rahman

1. Introduction and Rationale

Financial worry — the specific concern a household anticipates having the greatest difficulty meeting — is a compact indicator of household financial vulnerability, capturing whether a household is most preoccupied with everyday subsistence, a medical shock, education costs, old-age security, or business continuity. In the SAARC region, understanding this has direct relevance for financial-inclusion policy, microinsurance design, and social safety nets. Most existing evidence on this outcome comes from high-income countries, and prior South Asian studies have typically pooled respondents without an explicit country identifier or continuous age, constraining life-cycle analysis. This project proposes to use a dataset that, for the first time in this line of work, makes both available — allowing country-level and life-cycle effects to be modelled explicitly. Because the outcome is multinomial and unordered across six categories (Daily/Monthly Expenses, Medical Emergency, Education, Old Age, Business, Other/Unspecified), and predictor relationships may be non-linear, the study will benchmark a classical multinomial logit against SVM, Random Forest, and XGBoost, with SHAP used to interpret the best-performing model.

2. Brief Review of Literature

de Bruijn and Antonides (2020) and Magwegwe (2023) identify income, age, and both objective and subjective stressors as key drivers of financial worry. The Global Findex Database (Demirgüç-Kunt et al., 2022; Klapper et al., 2025) shows adults in developing countries worry most about everyday expenses and healthcare, alongside persistent digital-access gender gaps; Kumar and Pradhan (2024) and Mani (2016) find meaningful country-to-country variation in South Asia, motivating country as an explicit predictor. Wellalage and Locke (2017) document tighter credit constraints for female-owned SMEs, while the life-cycle hypothesis (Modigliani & Brumberg, 1954) motivates this study's age-related predictions. XGBoost (Chen & Guestrin, 2016) and SHAP (Lundberg & Lee, 2017) are now standard in applied financial-behaviour research, and Bergsma's (2013) bias-corrected Cramér's V will guide the bivariate analysis.

3. Objectives

General objective: to identify and explain the determinants of the greatest financial worry among households in SAARC countries. Specific objectives:

- Examine the distribution of financial worry and its demographic, economic, and digital-finance correlates.
- Test multicollinearity via correlation and VIF diagnostics prior to modelling.
- Estimate a multinomial logit and interpret Relative Risk Ratios.
- Benchmark the logit against SVM, Random Forest, and XGBoost on common performance metrics.
- Apply SHAP explainability to the best-performing model to rank key drivers of each worry category.
- Draw policy-relevant conclusions on financial-inclusion levers to reduce household vulnerability.

4. Data Source and Methodology

The analysis will use a cross-sectional sample of ~7,000 respondents from a Global Findex-style survey across India, Pakistan, Bangladesh, Sri Lanka, and Nepal (~43% India, ~14% each remaining country, reflecting regional population shares). A defined set of 26 raw variables spanning demographic, labor-market, financial-inclusion, digital-finance, and resilience domains will be encoded into the predictor matrix.

- Data cleaning, target recoding, and predictor construction, excluding variables that would cause target leakage.
- Exploratory and bivariate analysis (Chi-square, bias-corrected Cramér's V, Kruskal–Wallis for age).
- Multicollinearity diagnostics (correlation matrix, VIF), followed by multinomial logit estimation.
- SVM, Random Forest, and XGBoost training/evaluation on a common split (accuracy, macro-F1, weighted-F1, macro-AUC).
- SHAP explainability on the best-performing classifier.

Tools: Python 3.12 (pandas, NumPy, seaborn/matplotlib, SciPy, statsmodels, scikit-learn, XGBoost, shap).

5. Expected Outcome

A ranked understanding of the demographic, economic, digital-finance, and country-level correlates of each financial-worry category; a comparative assessment of classical versus machine-learning approaches; and policy-relevant recommendations — e.g., expanding mobile-money access or targeted microinsurance — to reduce household financial vulnerability in the SAARC region.

6. Proposed Work Plan (14 weeks, tentative)

Phase	Activities	Duration
1	Literature review, data acquisition, cleaning, and exploratory/bivariate analysis	Wks 1
2	Multicollinearity diagnostics and multinomial logistic regression	Wks 5–6
3	ML classifiers (SVM, Random Forest, XGBoost), benchmarking, and SHAP analysis	Wks 7–11
4	Report writing, supervisor review, revision, and final submission	Wks 12–14

References

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